

6.10 Package thermal characteristics

The maximum chip-junction temperature, $T_J \text{ max}$, in degrees Celsius, can be calculated using the following equation:

$$T_J \text{ max} = T_A \text{ max} + (P_D \text{ max} \times \Theta_{JA})$$

Where:

- $T_A \text{ max}$ is the maximum ambient temperature in °C.
 - Θ_{JA} is the package junction-to-ambient thermal resistance in °C/W.
 - $P_D \text{ max}$ is the sum of $P_{INT} \text{ max}$ and $P_{I/O} \text{ max}$:

$$P_D \text{ max} = P_{INT} \text{ max} + P_{I/O} \text{ max}$$
 - $P_{INT} \text{ max}$ is the product of I_{DD} and V_{DD} , expressed in Watts. This is the maximum chip internal power.
- $P_{I/O} \text{ max}$ represents the maximum power dissipation on output pins:

$$P_{I/O} \text{ max} = \sum (V_{OL} \times I_{OL}) + \sum ((V_{DDIOx} - V_{OH}) \times I_{OH})$$

taking into account the actual V_{OL}/I_{OL} and V_{OH}/I_{OH} of the I/Os at low and high level in the application.

Table 82. Package thermal characteristics

Symbol	Definition	Parameter	Value			Unit
			Junction-ambient Θ_{JA}	Junction-board Θ_{JB}	Junction-case Θ_{JC}	
Θ	Thermal Resistance	LQFP64	45.3	27.6	14.2	°C/W
		LQFP48	52.3	29.7	16.9	
		LQFP32	52.3	29.7	16.9	
		UFQFPN48	31.4	15.6	11.4	
		UFQFPN32	43.3	25.2	18.8	
		UFQFPN24	47.5	27.4	25.8	
		UFQFPN20	64.3	21.7	33.9	
		TSSOP20	78.9	52.4	28.4	

6.10.1 Reference documents

- JESD51-2 Integrated Circuits Thermal Test Method Environment Conditions - Natural Convection (Still Air) available on www.jedec.org.
- For information on thermal management, refer to application note "*Guidelines for thermal management on STM32 applications*" (AN5036) available on www.st.com.